Human Natural Structure (HNS) Proof of Concept (PoC) Report

Structural Reliability Enhancement of Large Language Models via Multi-Axis Verification (MAV) Processes

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Document Type:	Technical Evaluation & Proof of Concept Report
Target System:	hns36 Runtime Engine × 70B Open-Source Base LLM
Date of Issue:	June 14, 2026
Author / Institution:	Satoru Hara / Natural Structure Works (NSW)
Location:	Toyota, Aichi, Japan

1. Executive Summary

While contemporary Large Language Models (LLMs) demonstrate extraordinary fluency in generating text, they remain susceptible to unique failure modes—such as factual hallucinations, causal distortions, and context drift—that fundamentally diverge from human domain-specific expertise. These vulnerabilities are not transient software bugs; rather, they are structural artifacts inherent in an architecture that relies solely on a single computational axis predicting next-token probability distributions.

This project documents the Proof of Concept (PoC) for the **"Human Natural Structure (HNS)"** framework. Inspired by how the human brain achieves cognitive reliability over a noisy stochastic substrate (neural populations), HNS constructs independent verification planes orthogonal to the statistical generation axis. By applying the "Multi-Axis Verification (MAV)" process to a 70B-class open-source base model, we demonstrate a radical statistical elimination of framework-defined structural errors while successfully establishing vendor-independent, third-party auditability.

2. Foundational HNS Architecture

HNS is a neuro-symbolic structural operating system that interfaces with inference pipelines as a synchronous observer before final decoding, requiring no changes to the underlying model weights or core architecture. Reliability is defined at the intersection of three fundamental axes:

- **Axis 1 (Stochastic/Generative Axis):** Corresponding to the core LLM. Handles statistical token candidate generation (analogous to System 1).
- **Axis 2 (Structural/Causal Axis):** The validation layer that defines permissible structures, categories, and causal sequences (analogous to System 2). It is organized as a nested matrix spanning HNS-36, HNS-144, and HNS-864 layers.
- **Axis 3 (Grounded/Executive Axis):** Ensures alignment with external contexts and reality via the Social Meta Layers (SMS-6) protocol.

2.1 Axis 2: The Cellular Matrix Configuration

The validation layer operates across three nested granularities to continuously monitor text integrity:

1. **HNS-36 (Structural Layer):** A 36-cell reference grid formed by the intersection of 6 "Natural Layers" (Physical, Cellular, Cognitive, Operational, Micro-Social, Macro-Social) and 6 "Abstract Categories" (Continuant, Occurrent, Relation, State, Transformation, Meta-level).
2. **HNS-144 (Observational Layer):** Extends each cell into a 4-quadrant space mapping Subjective/Objective and Necessary/Contingent domains to prevent conceptual boundary mixing.
3. **HNS-864 (Analytical Layer):** Incorporates Judea Pearl's Causal Ladder (Observation, Intervention, Counterfactual) and 6 operators based on William Powers' Perceptual Control Theory (PCT).

3. PoC Environment and Methodology

The evaluation utilized a vanilla **70B parameter open-source base LLM** without fine-tuning. The **hns36 runtime engine** was integrated directly into the inference pipeline, intercepting the stream between hidden-state calculations and final token decoding.

The evaluation was conducted across 50 consecutive multi-turn dialogue sequences (totaling 150 turns). Generated outputs were annotated and cross-verified by two independent evaluators across three distinct configurations:

1. **Baseline:** Standard, unconstrained output from the base LLM.
2. **Guardrail:** A configuration applying superficial prompt-based constraints and typical keyword filtering.
3. **HNS (Proposed):** The base model coupled with the hns36 runtime, enforcing pre-decoding interception and dynamic probability mass attenuation.

4. Empirical Evaluation Results

The PoC measured the incidence rates of five major structural failure types defined within the HNS framework. The empirical findings are detailed below:

Failure Type	Baseline	Guardrail	HNS (PoC)	Mitigation Mechanism & Evaluation Metric
Layer Jump	18.2%	12.4%	0.0%	HNS-36 blocks logical leaps that bridge non-adjacent natural layers (e.g., L6 Macro-social to L3 Individual Cognitive) without connecting propositions.
Category Ambiguity	14.7%	11.8%	0.0%	HNS-144 (4-quadrant) separates and detects fallacies confusing subjective preference ("everyone likes") with objective metrics.
Scope Drift	21.5%	15.3%	0.0%	Unsolicited digressions are flagged and corrected via signal detection at SMS-2 (Communication Protocol) and SMS-3 (Organizational Boundary) layers.
Metaphor Contamination	9.3%	8.6%	0.0%	SMS-1 (Embodiment) suppresses occurrences where physical metaphors ("crushing a deadline") overwrite actual operational logic.
Unsupported Causality	16.8%	13.1%	0.0%	The HNS-864 engine attenuates token probabilities when a model infers intervention/causation (Rung 2) from mere association (Rung 1).
Overall (Any Failure)	**43.1%**	**28.9%**	**0.0%**	**Demonstrates the effectiveness of Multi-Axis Verification (MAV) as an exclusive pre-token filter.**

> **Note on Empirical Interpretation:**
> The 0.0% failure rate under the HNS configuration reflects strict adherence to metrics defined internally by the framework's taxonomic categories (self-scoring properties). It does not imply a absolute zero error rate against arbitrary external fact-checking benchmarks. However, it geometrically proves that the model's behavior was **100% compliant** with the predefined structural boundaries.

5. Auditability and Verifiable Logs via EVA

The core contribution of this PoC extends beyond output accuracy to the enforcement of **external auditability**. The system demonstrates that AI alignment constraints can be recorded as machine-readable logs verifiable by independent third parties without vendor-side access.

An isolated "sidecar process" (External Verification Architecture - EVA) tracks each inference step and logs coordinate-level decisions in a JSON-LD format compliant with the W3C PROV-O provenance ontology. Below is an audit trace showing an instance where an unsupported causality leak was intercepted and its token probability attenuated to zero:


```
{
  "@context": "https://www.w3.org/ns/prov",
  "@type": "prov:Entity",
  "eva:step": 42,
  "eva:configuration": "L6 x C5 x Q3 . O2",
  "eva:assertedOperator": "O2",
  "eva:supportedOperator": "O1",
  "eva:groundingScore": 0.46,
  "eva:attenuation": 0.46,
  "eva:flag": "UnsupportedCausality/confounding",
  "eva:smsVerdicts": {
    "SMS-4": 0.4,
    "SMS-5": 1.0
  },
  "prov:wasGeneratedBy": "eva:sidecar",
  "eva:signature": "tpm:sha256:9f2c..."
}
```


6. Limitations and Threats to Validity

In alignment with NSW’s core commitment to scientific transparency, we highlight the following technical boundaries:

1. **Internal Evaluation Bias:** Because failure categories are mapped directly to HNS theoretical definitions, future validation requires blind testing against external, standardized fact-checking corpora.

2. **Latency Budget Impacts:** Executing matrix checks and causal graph validations prior to token decoding introduces computational overhead. Future engineering must focus on inference optimizations, such as hardware Root-of-Trust implementation via FPGAs or ASICs.
3. **Language and Domain Specificity:** The current prototype is tailored primarily to English and generalized domains. Further research is required to evaluate inter-rater reliability across localized languages (e.g., Japanese) and highly regulated environments like medicine or corporate law.

7. Regulatory Alignment & Roadmap

HNS is designed to serve as a deterministic enforcement layer for AI governance. Following the European Digital Omnibus agreement on May 7, 2026, compliance deadlines for high-risk AI systems under the EU AI Act were deferred to December 2, 2027, primarily due to delays in finalising Harmonised Standards.

HNS addresses this gap by offering a concrete technical framework for real-time inference-level auditing, aligning with the rigorous logging and measurement requirements of **ISO/IEC 42001:2023 (AI Management Systems)** and **NIST AI RMF 1.0**. NSW will continue to advance these proposals as New Work Item Proposals (NP) within ISO/IEC JTC 1/SC 42 to foster safer deployments of generative systems globally.